

Yige Yan

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Research & Technical Summary

Ageing and Neurodegeneration · Disentanglement · Contrast-free Cardiac MRI

I recover disease signal that routine clinical data misses or confounds, and test whether models learn that signal or a shortcut, using multimodal, vision–language, foundation and state-space models. Seeking research and applied-science internships in medical AI, computer vision, multimodal and trustworthy AI.

Education

Nanyang Technological University

Singapore

Ph.D. in Computer Science and Engineering, GPA: 4.67/5.00

Aug 2023 – Present

Main supervisor: Prof. Jagath C. Rajapakse (IEEE Fellow), College of Computing and Data Science.

Co-supervisor: Assoc. Prof. Nagaendran Kandiah, Lee Kong Chian School of Medicine.

Expected thesis submission: first half of 2027.

Nanyang Technological University

Singapore

M.Sc. in Computer Control and Automation, GPA: 4.44/5.00

Aug 2022 – Jun 2023

Supervisor: Prof. Wee Peng Tay, School of Electrical and Electronic Engineering.

Hohai University

Nanjing, China

B.Eng. in Computer Science, GPA: 4.48/5.00

Sep 2017 – Jun 2021

Supervisor: Dr. Xiaoqin Zhou, College of Artificial Intelligence and Automation.

Publications & Manuscripts

1. **Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cardiac Magnetic Resonance.” *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2024. **Early Accept.**
2. **Yige Yan**, Jun Cheng, Xulei Yang, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Dual Correlation-Aware Mamba for Microvascular Obstruction Identification in Non-contrast Cine Cardiac Magnetic Resonance.” *MICCAI*, 2025. **Early Accept.**
3. **Yige Yan**, Yi Hao Chan, Jun Cheng, Qian Hui Chew, Kang Sim, and Jagath C. Rajapakse. “Latent Pathology–Age Decoupling in Functional Connectivity Graphs for Multi-task Schizophrenia Assessment.” *MICCAI*, 2026. **Early Accept.**
4. **Yige Yan**, Jing Xia, Yi Hao Chan, Thomas Welton, and Jagath C. Rajapakse. “Decoupling Age Effects from Parkinson’s Disease Functional Connectivity.” *Medical Image Analysis*. **Under review.**
5. **Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Spatiotemporal Adaptive Mamba with Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cine Cardiac MRI.” *Neurocomputing*. **Under review.**
6. **Yige Yan**, et al. “When Should an Image Swap Change a Vision–Language Model’s Answer? Evaluating Answer Preservation and Correct Updating.” *International Conference on Learning Representations (ICLR)*, 2027. **Under review.** [TODO: replace “et al.” with the final author list]
7. **Yige Yan** and Jagath C. Rajapakse, for the Alzheimer’s Disease Neuroimaging Initiative. “Clinical Evidence Reconfigures Age Reliance in Language Models Assessing Alzheimer’s Disease.” **Prepared for submission to npj Digital Medicine.**
8. **Yige Yan**, et al. “Temporal Variation Proxy and Staged Proposal–Support Integration for Contrast-Free MVO Segmentation in Cine CMR.” **Prepared for submission to IEEE Transactions on Medical Imaging.**
9. **Yige Yan**, et al. “Auditing Named Knowledge Interfaces: Regularization, Responsiveness, and

Predictive Utility.” *Prepared for submission to Knowledge-Based Systems.*

Conference Presentations & Posters

- *MICCAI 2026*, Strasbourg, France, Sep–Oct 2026. Poster presentation.
- *MICCAI 2025*, Daejeon, Republic of Korea, Sep 2025. Poster presentation.
- *IEEE EMBS – Duke-NUS – HealthTEC.SG – NUS iHealthtech Forum 2025: AI in Healthcare and MedTech*, Singapore, Apr 2025. Poster presentation.
- *MICCAI 2024*, Marrakesh, Morocco, Oct 2024. Poster presentation.
- *Singapore Healthcare Engineering Conference (SHEC) 2024*, IEEE EMB Singapore Chapter, Singapore, Oct 2024. Poster presentation.
- *AI Health Summit 2023*, SingHealth, Ministry of Health and A*STAR, Singapore, Nov 2023. Poster presentation, [TODO: OCTA poster title].

Research Experience

Nanyang Technological University

Ph.D. Researcher, College of Computing and Data Science

Singapore

Aug 2023 – Present

- **Contrast-free cardiac MRI.** The sequence that defines microvascular obstruction requires gadolinium and is absent from routine cine acquisition. Built a multi-year line recovering the finding from cine alone, from coarse-grained supervision under imperfect LGE-to-cine alignment through motion modeling and spatiotemporal state-space learning to pixel-level segmentation.
- **Confounder-free functional connectivity.** Disease signal is present but dominated by age-associated variation. Designed graph-based representation learning that disentangles the two and reintroduces age as controlled clinical context, across schizophrenia, Parkinson’s disease and Alzheimer’s disease.

Institute for Infocomm Research (I²R), A*STAR

Research Student Attachment, Medical Imaging

Singapore

Aug 2023 – Present

- Microvascular obstruction identification from non-contrast cine cardiac MRI, with collaborators across A*STAR, NTU, Duke-NUS Medical School and the National Heart Centre Singapore.

Institute for Infocomm Research (I²R), A*STAR

Research Student Attachment, OCTA Denoising

Singapore

Mar 2023 – Jun 2023

- Deep-learning denoising and enhancement for optical coherence tomography angiography (OCTA), with collaborators across A*STAR, NTU and the Singapore National Eye Centre.

Research Grants & Proposal Experience

Singapore MOE Academic Research Fund (AcRF) Tier 2

Primary proposal writer and research-plan developer

In preparation, Sep 2026

Faculty PI: Prof. Jagath C. Rajapakse

“Separating and Contextualizing Age for Reliable Assessment and Prognosis of Neurodegenerative Disease.”
Drafted the research plan and preliminary-study design, extending my doctoral research on pathology–age decoupling into a multi-project research program.

Teaching Experience

Nanyang Technological University

Teaching Assistant, College of Computing and Data Science

Singapore

2024 – Present

- **SC4001 Neural Networks and Deep Learning**, lead teaching assistant. Coordinated the teaching-assistant group and authored assignment problems. Jan–Apr 2025, Aug–Nov 2025, Aug 2026–present.
- **SC1013 Physics for Computing**, lab supervisor. Aug–Nov 2025.

- **SC1006 Computer Organisation & Architecture**, lab supervisor. Feb–Mar 2025.
- **SC2079 Multidisciplinary Design Project**, lab supervisor. Aug–Oct 2024.
- **SC2008 Computer Network**, lab supervisor. Feb–Apr 2024.

Research Mentoring

Nanyang Technological University

Mentor, Undergraduate Final-Year Projects (FYP)

Singapore

2025 – Present

- AY2026/27. “Predicting Multiple Cognitive Functions Simultaneously from Brain Functional Connectivity via an Interpretable Multi-Task Learning Network.”
- AY2026/27. Age-conditioned Riemannian residual modeling of schizophrenia functional connectomes.
- AY2025/26. “Explainable AI for Functional MRI Deep Learning Model Analysis.”
- AY2025/26. “Multi-Modal Functional MRI Fusion for Fluid Intelligence Prediction.”
- AY2025/26. Graph state-space modeling for Alzheimer’s disease neuroimaging.

Academic Service

Reviewer: MICCAI; IEEE Transactions on Neural Networks and Learning Systems (TNNLS); IEEE Transactions on Image Processing (TIP).

Honors, Awards & Scholarships

- **MICCAI 2024 Registration and Travel Award.**
- **Nanyang Technological University Research Scholarship (RSS).**
- **Outstanding Undergraduate Graduation Project**, Hohai University, 2021.
- **Outstanding Student Award**, Hohai University.
- **Academic Merit Scholarship**, Hohai University.

Technical Skills

Programming & Frameworks: Python, PyTorch, PyTorch Geometric, C++, C, NumPy, SciPy, scikit-learn.

Domains: Medical image analysis, cardiac MRI, fMRI, neuroimaging, OCTA, medical multimodal AI.

Languages: Chinese (native), English (fluent), Japanese (proficient).